

Vinal DalcY Dsouza

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Available from May 10, 2026 onwards

Education

Northeastern University, Oakland, CA Sep 2024 – Present
Master of Science in Computer Science GPA: 4.00 / 4.00

Relevant Coursework: DBMS, Algorithms, Cloud Computing, Design Paradigms

NMAM Institute of Technology, Nitte, India Aug 2015 – Jun 2019
Bachelor of Engineering in Information Science GPA: 9.09 / 10

Relevant Coursework: AI, Data Mining, Data Structures, BI, Engineering Management

Experience

Accenture, Bengaluru, India Mar 2022 – Jun 2024
Data Engineering, Management and Governance Senior Analyst

- Led continuous improvement initiatives across BODS, EDW, and DataStage, delivering 10+ automations and saving 70+ FTEs annually.
- Designed batch and near real-time ETL pipelines using IBM DataStage to ingest MQ, XML, CSV, and Oracle data into ingestion, exploitation, and reporting layers.
- Implemented CDC and SCD-based data models to support incremental loads and historical accuracy.
- Reduced report generation time from 30 minutes to 15 seconds, saving 24 FTEs annually.
- Orchestrated and scheduled 50+ DataStage jobs using Tivoli Workload Scheduler, ensuring SLA compliance.
- Reduced L2 production support dependency by 50% via shift-left strategy for monthly credit risk validations.
- Debugged and resolved critical production issues using Python and SQL, improving system reliability.
- Collaborated with distributed teams across regions/business units to troubleshoot production issues.

Infosys, Chennai, India Sep 2019 – Mar 2022
Senior Systems Engineer

- Migrated 8+ COBOL-based ETL pipelines to IBM DataStage with 98% data accuracy and Autosys orchestration.
- Improved ETL workflow efficiency by 10% through reusable DataStage job design.
- Built complex transformations on large-scale financial datasets to support downstream reporting systems.
- Developed data mappings and Unix shell scripts, reducing manual housekeeping by 8 hours annually.
- Tuned critical SQL queries by optimizing execution plans and adding appropriate indexes on Oracle tables.

Projects

FMCG Delta Medallion Pipeline [Link](#) Dec 2025 – Jan 2026
PySpark, Delta Lake, Databricks Lakeflow

- Architected an automated medallion lakehouse with Databricks Workflows orchestrating tasks with parallel execution and error handling.
- Designed star schema and implemented incremental and full-load pipelines using Delta Lake MERGE operations and 10+ data quality checks across dimensions and facts.

Flight Delay Prediction (AWS SageMaker) [Link](#) Mar 2025 – Apr 2025
AWS SageMaker, XGBoost, Python

- Trained an XGBoost model on 580K+ flight records, improving delay detection sensitivity from 0.2% to 11.6%.
- Built end-to-end preprocessing and feature engineering pipelines including one-hot encoding and ZIP-to-CSV ingestion.

Microservices and CI/CD Pipeline [Certificate](#) Feb 2025 – Mar 2025
Node.js, Docker, AWS ECS Fargate, CI/CD

- Decomposed a monolithic Node.js application into Dockerized microservices deployed on AWS ECS Fargate.
- Implemented CI/CD pipelines using CodePipeline, CodeDeploy, IAM roles, and S3 artifact storage with blue-green deployments.

Skills

Languages: Python, C, C++, JavaScript, SQL

Data & ML: NumPy, Pandas, Matplotlib, Apache Spark, AWS Glue, Databricks, DataStage, SSIS, Power BI

Databases: PostgreSQL (pgvector), MySQL, MongoDB, Redis, Oracle

Cloud & DevOps: AWS (S3, EMR, Glue, Lambda, SageMaker, ECS, IAM, CodePipeline, CodeDeploy), Docker, Firebase

Workflow & Version Control: Apache Kafka, Autosys, Tivoli Workload Scheduler, Git, GitHub, Bitbucket